

```

> use gestion_academica2;
< switched to db gestion_academica2
> db.createCollection("docentes_validados", {
  validator: {
    $jsonSchema: {
      bsonType: "object",
      required: ["nombres","apellidos","dui","email","telefono","especialidad","fecha_contratacion","activo"],
      properties: {
        nombres: { bsonType: "string", minLength: 2, maxLength: 60 },
        apellidos: { bsonType: "string", minLength: 2, maxLength: 60 },
        dui: { bsonType: "string", pattern: "^([0-9]{8}-[0-9]{1})$" },
        email: { bsonType: "string", pattern: "^([a-zA-Z0-9._%+-]+@institucion\\.edu\\.sv)$" },
        telefono: { bsonType: "string", pattern: "^([0-9]{4}-[0-9]{4})$" },
        especialidad: { bsonType: "string", minLength: 3, maxLength: 80 },
        fecha_contratacion: { bsonType: "date" },
        activo: { bsonType: "bool" }
      }
    }
  },
  validationLevel: "strict",
  validationAction: "error"
})
< { ok: 1 }
gestion_academica2>

```

```

gestion_academica2> db.createCollection("estudiantes_validados", {
  validator: {
    $jsonSchema: {
      bsonType: "object",
      required: ["carnet","nombres","apellidos","fecha_nacimiento","genero","email","telefono","direccion","carrera_codigo","ciclo_ingreso","activo"],
      properties: {
        carnet: { bsonType: "string", pattern: "^2026-[A-Z]{2,3}-[0-9]{3}$" },
        nombres: { bsonType: "string", minLength: 2, maxLength: 60 },
        apellidos: { bsonType: "string", minLength: 2, maxLength: 60 },
        fecha_nacimiento: { bsonType: "date" },
        genero: { enum: ["Masculino","Femenino"] },
        email: { bsonType: "string", pattern: "^([a-zA-Z0-9._%+-]+@alumno\\.edu\\.sv)$" },
        telefono: { bsonType: "string", pattern: "^([0-9]{4}-[0-9]{4})$" },
        direccion: {
          bsonType: "object",
          required: ["departamento","municipio","detalle"],
          properties: {
            departamento: { bsonType: "string", minLength: 2 },
            municipio: { bsonType: "string", minLength: 2 },
            detalle: { bsonType: "string", minLength: 5, maxLength: 150 }
          }
        },
        carrera_codigo: { bsonType: "string", minLength: 2, maxLength: 10 },
        ciclo_ingreso: { bsonType: "string", pattern: "^([I|II]{1,2}-[0-9]{4})$" },
        materias_aprobadas: { bsonType: "array", items: { bsonType: "string" } },
        activo: { bsonType: "bool" }
      }
    }
  },
  validationLevel: "strict",
  validationAction: "error"
})

```

```

db.createCollection("materias_validadas", {
  validator: {
    $jsonSchema: {
      bsonType: "object",
      required: ["codigo","nombre","uv","ciclo","area","activo"],
      properties: {
        codigo: { bsonType: "string", pattern: "^[A-Z]{2,5}[0-9]{3}$" },
        nombre: { bsonType: "string", minLength: 5, maxLength: 100 },
        uv: { bsonType: "int", minimum: 1, maximum: 6 },
        ciclo: { bsonType: "int", minimum: 1, maximum: 10 },
        area: { enum: ["Base de Datos","Programación","Ingeniería de Software","Gastronomía","Logística","Aduanas","Turismo"] },
        prerequisitos: { bsonType: "array", items: { bsonType: "string" } },
        activo: { bsonType: "bool" }
      }
    }
  },
  validationLevel: "strict",
  validationAction: "error"
})
{ ok: 1 }
gestion_academica2>

```

```

gestion_academica2> db.createCollection("inscripciones_validadas", {
  validator: {
    $jsonSchema: {
      bsonType: "object",
      required: ["carnet_estudiante","codigo_grupo","fecha_inscripcion","estado"],
      properties: {
        carnet_estudiante: { bsonType: "string" },
        codigo_grupo: { bsonType: "string" },
        fecha_inscripcion: { bsonType: "date" },
        estado: { enum: ["Inscrito","Retirado","Aprobado","Reprobado"] }
      }
    }
  },
  validationLevel: "strict",
  validationAction: "error"
})

```

```

gestion_academica2> db.createCollection("evaluaciones_validadas", {
  validator: {
    $jsonSchema: {
      bsonType: "object",
      required: ["codigo_evaluacion","codigo_grupo","nombre_evaluacion","porcentaje","fecha","activo"],
      properties: {
        codigo_evaluacion: { bsonType: "string", pattern: "^EV-[A-Z0-9]{1}-[0-9]{3}$" },
        codigo_grupo: { bsonType: "string" },
        nombre_evaluacion: { enum: ["Primer Parcial","Segundo Parcial","Proyecto Final","Laboratorio","Investigación","Exposición","Práctica Guiada","Proyecto Práctico"] },
        porcentaje: { bsonType: "int", minimum: 1, maximum: 100 },
        fecha: { bsonType: "date" },
        activo: { bsonType: "bool" }
      }
    }
  },
  validationLevel: "strict",
  validationAction: "error"
})

```

```

gestion_academica2> db.createCollection("calificaciones_validadas", {
  validator: {
    $jsonSchema: {
      bsonType: "object",
      required: ["codigo_evaluacion","carnet_estudiante","nota","fecha_registro"],
      properties: {
        codigo_evaluacion: { bsonType: "string" },
        carnet_estudiante: { bsonType: "string" },
        nota: { bsonType: ["double","int"], minimum: 0, maximum: 10 },
        observacion: { bsonType: "string", maxLength: 200 },
        fecha_registro: { bsonType: "date" }
      }
    }
  },
  validationLevel: "strict",
  validationAction: "error"
})
|

```

```

> use('admin')

db.createUser(
  {
    user: "admin",
    pwd: "Admin123*",
    roles: [
      {
        role: "root",
        db: "admin"
      }
    ]
  }
)
< { ok: 1 }
admin> |

```

```

test> use gestion_academica2
switched to db gestion_academica2
gestion_academica2> db.createUser({
|   user: "docente",
|   pwd: "Docente123*",
|   roles: [
|     {
|       role: "readWrite",
|       db: "gestion_academica2"
|     }
|   ]
| })
|
{ ok: 1 }
gestion_academica2>

```

Crear un usuario llamado coordinador con permisos dbAdmin sobre la base de datos gestion_academica.

```

test> use gestion_academica2
switched to db gestion_academica2
gestion_academica2> db.createUser({
|   user: "coordinador",
|   pwd: "Coord123*",
|   roles: [
|     {
|       role: "dbAdmin",
|       db: "gestion_academica2"
|     }
|   ]
| })
|
{ ok: 1 }
gestion_academica2>

```

Crear un usuario llamado auditor con permisos únicamente de lectura.

```

gestion_academica2> db.createUser({
|   user: "auditor",
|   pwd: "Audit123*",
|   roles: [ { role: "read", db: "gestion_academica"
|   })
| })
{ ok: 1 }
gestion_academica2>

```

Intentar insertar información utilizando un usuario con permisos read.

```
~ ● x mongosh -u docente -p Docente123* --authenticationDatabase gestion_academica2
Current Mongosh Log ID: 6a34cb32aa0f1790ca9df8a2
Connecting to:      mongodb://<credentials>@127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&authSource=gestion_academica2&appName=mongosh+2.8.3
Using MongoDB:      7.0.37
Using Mongosh:      2.8.3

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

test> use gestion_academica2
switched to db gestion_academica2
gestion_academica2> db.estudiantes.insertOne({ nombre: "Carlos" })
|
{
  acknowledged: true,
  insertedId: ObjectId('6a34cb3faa0f1790ca9df8a3')
}
gestion_academica2> █
```

```
gestion_academica2> exit;
~ ● → mongosh -u auditor -p Auditor123* --authenticationDatabase gestion_academica2
Current Mongosh Log ID: 6a34cb67b040cb008e9df8a2
Connecting to:      mongodb://<credentials>@127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&authSource=gestion_academica2&appName=mongosh+2.8.3
Using MongoDB:      7.0.37
Using Mongosh:      2.8.3

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

test> use gestion_academica2
switched to db gestion_academica2
gestion_academica2> db.estudiantes.insertOne({ nombre: "Prueba" })
|
MongoServerError[Unauthorized]: not authorized on gestion_academica2 to execute command { insert:
"estudiantes", documents: [ { nombre: "Prueba", _id: ObjectId('6a34cb6fb040cb008e9df8a3') } ], ordered: true, lsid: { id: UUID("c4f04ea8-688f-49e6-ad63-4abab5924909") }, $db: "gestion_academica2"
}
gestion_academica2> █
```


Investigue como realizar copias de seguridad y como poder restaurarlas:

```
~ ➔ mkdir backups
mkdir: created directory 'backups'
~ ➔ sudo chown -R srmochi:srmochi backups/
~ ➔ sudo chmod -R 755 backups/
~ ➔ mongodump --db gestion_academica2 \
--out /home/srmochi/backups \
--username admin \
--password Admin123* \
--authenticationDatabase admin
2026-06-18T22:59:57.204-0600 writing 'gestion_academica2.estudiantes' to '/home/srmochi/backups/gestion_academica2/estudiantes.bson'
2026-06-18T22:59:57.204-0600 writing 'gestion_academica2.docentes_validados' to '/home/srmochi/backups/gestion_academica2/docentes_validados.bson'
2026-06-18T22:59:57.204-0600 writing 'gestion_academica2.calificaciones_validadas' to '/home/srmochi/backups/gestion_academica2/calificaciones_validadas.bson'
2026-06-18T22:59:57.204-0600 writing 'gestion_academica2.estudiantes_validados' to '/home/srmochi/backups/gestion_academica2/estudiantes_validados.bson'
2026-06-18T22:59:57.213-0600 done dumping 'gestion_academica2.estudiantes' (1 documents)
2026-06-18T22:59:57.213-0600 writing 'gestion_academica2.inscripciones_validadas' to '/home/srmochi/backups/gestion_academica2/inscripciones_validadas.bson'
2026-06-18T22:59:57.214-0600 done dumping 'gestion_academica2.calificaciones_validadas' (0 documents)
2026-06-18T22:59:57.214-0600 writing 'gestion_academica2.evaluaciones_validadas' to '/home/srmochi/backups/gestion_academica2/evaluaciones_validadas.bson'
2026-06-18T22:59:57.216-0600 done dumping 'gestion_academica2.estudiantes_validados' (0 documents)
2026-06-18T22:59:57.216-0600 writing 'gestion_academica2.materias_validadas' to '/home/srmochi/backups/gestion_academica2/materias_validadas.bson'
2026-06-18T22:59:57.217-0600 done dumping 'gestion_academica2.docentes_validados' (0 documents)
2026-06-18T22:59:57.217-0600 writing 'gestion_academica2.carreras' to '/home/srmochi/backups/gestion_academica2/carreras.bson'
2026-06-18T22:59:57.219-0600 done dumping 'gestion_academica2.inscripciones_validadas' (0 documents)
2026-06-18T22:59:57.221-0600 done dumping 'gestion_academica2.evaluaciones_validadas' (0 documents)
2026-06-18T22:59:57.222-0600 done dumping 'gestion_academica2.carreras' (0 documents)
2026-06-18T22:59:57.224-0600 done dumping 'gestion_academica2.materias_validadas' (0 documents)
```

para restaurar es:

```
mongorestore --db gestion_academica2
/home/srmochi/backups/gestion_academica2 --username admin --
password Admin123* --authenticationDatabase admin
```

```
~ ➔ mongodump --db biblioteca_estrella_de_la_mañana \
--out /home/srmochi/backups \
--username admin \
--password Admin123* \
--authenticationDatabase admin
2026-06-18T23:13:16.332-0600 writing 'biblioteca_estrella_de_la_mañana.autores' to '/home/srmochi/backups/biblioteca_estrella_de_la_mañana/autores.bson'
2026-06-18T23:13:16.332-0600 writing 'biblioteca_estrella_de_la_mañana.usuarios' to '/home/srmochi/backups/biblioteca_estrella_de_la_mañana/usuarios.bson'
2026-06-18T23:13:16.332-0600 writing 'biblioteca_estrella_de_la_mañana.prestamos' to '/home/srmochi/backups/biblioteca_estrella_de_la_mañana/prestamos.bson'
2026-06-18T23:13:16.333-0600 writing 'biblioteca_estrella_de_la_mañana.libros' to '/home/srmochi/backups/biblioteca_estrella_de_la_mañana/libros.bson'
2026-06-18T23:13:16.342-0600 done dumping 'biblioteca_estrella_de_la_mañana.autores' (10 documents)
2026-06-18T23:13:16.344-0600 done dumping 'biblioteca_estrella_de_la_mañana.usuarios' (10 documents)
2026-06-18T23:13:16.349-0600 done dumping 'biblioteca_estrella_de_la_mañana.libros' (10 documents)
2026-06-18T23:13:16.350-0600 done dumping 'biblioteca_estrella_de_la_mañana.prestamos' (10 documents)
```

Investigue el comando para todos los usuarios creados en MongoDB.

```
use gestion_academica2
db.getUsers()
```

```

test> use gestion_academica2
switched to db gestion_academica2
gestion_academica2> db.getUsers()
{
  users: [
    {
      _id: 'gestion_academica2.auditor',
      userId: UUID('59d9a1cf-c887-4bf4-920d-0eaffbf254ce'),
      user: 'auditor',
      db: 'gestion_academica2',
      roles: [ { role: 'read', db: 'gestion_academica' } ],
      mechanisms: [ 'SCRAM-SHA-1', 'SCRAM-SHA-256' ]
    },
    {
      _id: 'gestion_academica2.coordinador',
      userId: UUID('144365bc-962c-492e-9070-f3748db3ff19'),
      user: 'coordinador',
      db: 'gestion_academica2',
      roles: [ { role: 'dbAdmin', db: 'gestion_academica2' } ],
      mechanisms: [ 'SCRAM-SHA-1', 'SCRAM-SHA-256' ]
    },
    {
      _id: 'gestion_academica2.docente',
      userId: UUID('fb2ef103-ab9a-482a-94e0-10f79cf35e22'),
      user: 'docente',
      db: 'gestion_academica2',
      roles: [ { role: 'readWrite', db: 'gestion_academica2' } ],
      mechanisms: [ 'SCRAM-SHA-1', 'SCRAM-SHA-256' ]
    }
  ],
  ok: 1
}
gestion_academica2>

```

Investigue cómo poder crear roles personalizados y cómo se pueden asignar a nuevos usuarios.

```

gestion_academica2> db.createRole({
|   role: "rol_catalogador",
|   privileges: [
|     { resource: { db: "gestion_academica2", collection: "materias" }, actions: ["find","insert"] }
|   ],
|   roles: []
| })
{ ok: 1 }

```

```
gestion_academica2> db.createUser({
|   user: "catalogador",
|   pwd: "Cat123*",
|   roles: [ { role: "rol_catalogador", db: "gestion_academica2" } ]
| })
|
{ ok: 1 }
gestion_academica2> █
```

Investigue cómo actualizar las credenciales y privilegios de un usuario.

```
gestion_academica2> db.updateUser("coordinador", {
|   roles: [ { role: "readWrite", db: "gestion_academica2" } ]
| })
|
{ ok: 1 }
gestion_academica2> █
```